Automatic Hierarchical Summarization of Scientific Texts

Fedor Alexandrovich Sobolevsky
Scientific supervisor — D. Sc. Konstantin Vyacheslavovich Vorontsov

Moscow Institute of Physics and Technology

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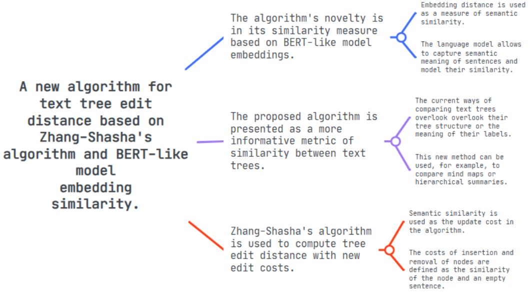
Problem Statement

A **hierarchical summary** is a summary of a document in the form of a **text tree** — a tree $T = (V, E)$, where $E \subset V^2$ and for each $v \in V$ a text $s(v) \in \mathcal{S}$ is defined.

Hierarchical summarization problem: Find a mapping $f : D \mapsto T$ that constructs a hierarchical summary $T \in \mathcal{T}$ from a document D, minimally different from a reference summary T^* of D constructed by an expert:

$$\rho(f(D), T^*) \longrightarrow \min_f .$$



An example of a hierarchical summary of this study

Problem Statement

The following requirements for hierarchical summaries are introduced:

1. **Completeness:** a hierarchical summary, expanded to an arbitrary depth, is a complete summary of the source document.
2. **Conciseness:** a hierarchical summary, expanded to an arbitrary depth, should be concise, i. e. it should present information from the source document in as brief of a way as possible without losing any relevant information.
3. **Coherence:** a hierarchical summary, expanded to an arbitrary depth, should be coherent when read in depth-first order.

Baseline metric — **TTED**[a].

Modifications:

1. Normalization[b]:

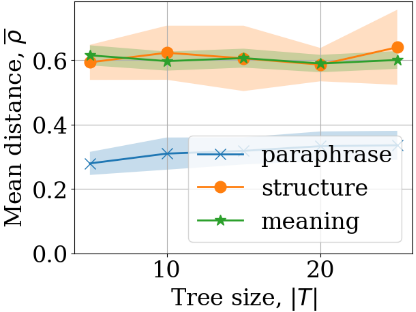
$$\mathrm{nTTED}(T_1, T_2) = \frac{2 \cdot \mathrm{TTED}(T_1, T_2)}{|T_1| + |T_2| + \mathrm{TTED}(T_1, T_2)};$$

2. Comparison at depth k:

$$\mathrm{TTED@}k(T_1, T_2) = \mathrm{TTED}(T_1@k, T_2@k),$$

[a] *Sobolevsky, Fedor and Vorontsov, Konstantin.* Text Tree Edit Distance: A Language Model-Based Metric for Text Hierarchies (2025)

[b] *Li, Yujian and Chenguang, Zhang.* A metric normalization of tree edit distance



Scores by nTTED are tree size-agnostic yet still reflect significant differences the most

Method: LLM Generation

The method for automatic hierarchical summary generation in this study is by using **large language models (LLMs)**.

Proof-of-concept experiment — comparison of LLM mind map generation against SoTA graph-based methods[a]. Similarity function based on R — ROUGE (as in other studies):

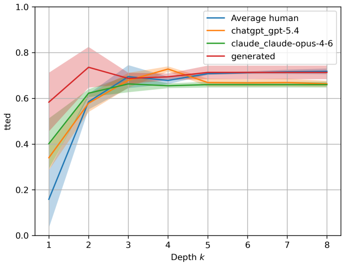
$$\mathrm{Sim}(T, T') = \max_{P \subset E \times E'} \sum_{(e,e') \in P} \sum_{i=0,1} \mathrm{R}(e_i, e'_i).$$

[a] *Zhang Zhuowei, Hu Mengting, Bai Yinhao, and Zhang Zhen.* Coreference Graph Guidance for Mind-Map Generation (2024)

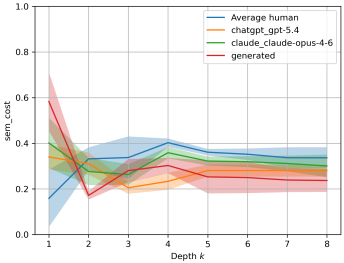
Models	**Sim**
Random	28.77
MRDMF	34.47
EMGN	42.73
CMGN	44.27
Qwen2.5-3B-Instruct (1-shot)	41.58
Qwen3-4B-Instruct (1-shot)	**46.83**

Mind map generation results. Even a small modern LLM can outperform graph-based methods with good prompting.

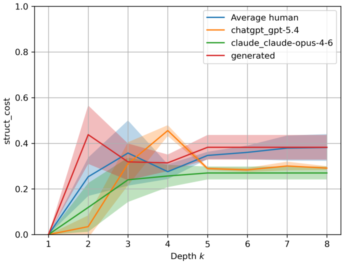
LLM Evaluation



TTED



Semantic component of TTED



Structural component of TTED

The results of LLM evaluation on a human hierarchical summary benchmark show that difference between LLM generated hierarchical summaries and references is on average not higher than between references.

Main Results

1. The requirements for a quality hierarchical summary have been formalized.
2. TTED modifications have been proposed for more flexible and interpretable summary comparison.
3. Small LLMs have been shown to outperform SoTA methods for mind map generation.
4. An evaluation framework has been built for the task of hierarchical summarization.
5. First results have been acquired for LLM evaluation on a human hierarchical summary benchmark.

References

- *Sobolevsky, F. and Vorontsov, K.* 2025. Text tree edit distance: A language model-based metric for text hierarchies. In 2025 IEEE XVII International Scientific and Technical Conference on Actual Problems of Electronic Instrument Engineering (APEIE), pages 1–5. IEEE.
- *Zhang, Z., Hu, M., et al.* 2024. Coreference Graph Guidance for Mind-Map Generation // Proceedings of the AAAI Conference on Artificial Intelligence. — Vol. 38. — P. 19623–19631.
- *Li, Y. and Chenguang, Z.* 2011. A metric normalization of tree edit distance. Frontiers of Computer Science in China, 5(1), pp. 119-125.